Assignment 4

ASTR4004

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1 Introduction

In this assignment I am to create a regression neural network to fit to an arbitrary function of form: $y = mx + b$, given two inputs x_1 and x_2. The model uses a feed forward method to calculate the hidden layers, gradients and outputs, and a MSE Loss function was used for the losses. Two different activation functions have been used: within the hidden layer a sigmoid is applied, while a reLU activation function has been applied to the output to not lose information on the gradients. Inspiration and process have been inspired and followed from week 10 lecture slides specifically slides 22-30.

2 Network Structure

The structure of the network which was followed has been drawn in Fig.1 - any understanding needed of subscripts used in the calculations section please refer to this image.

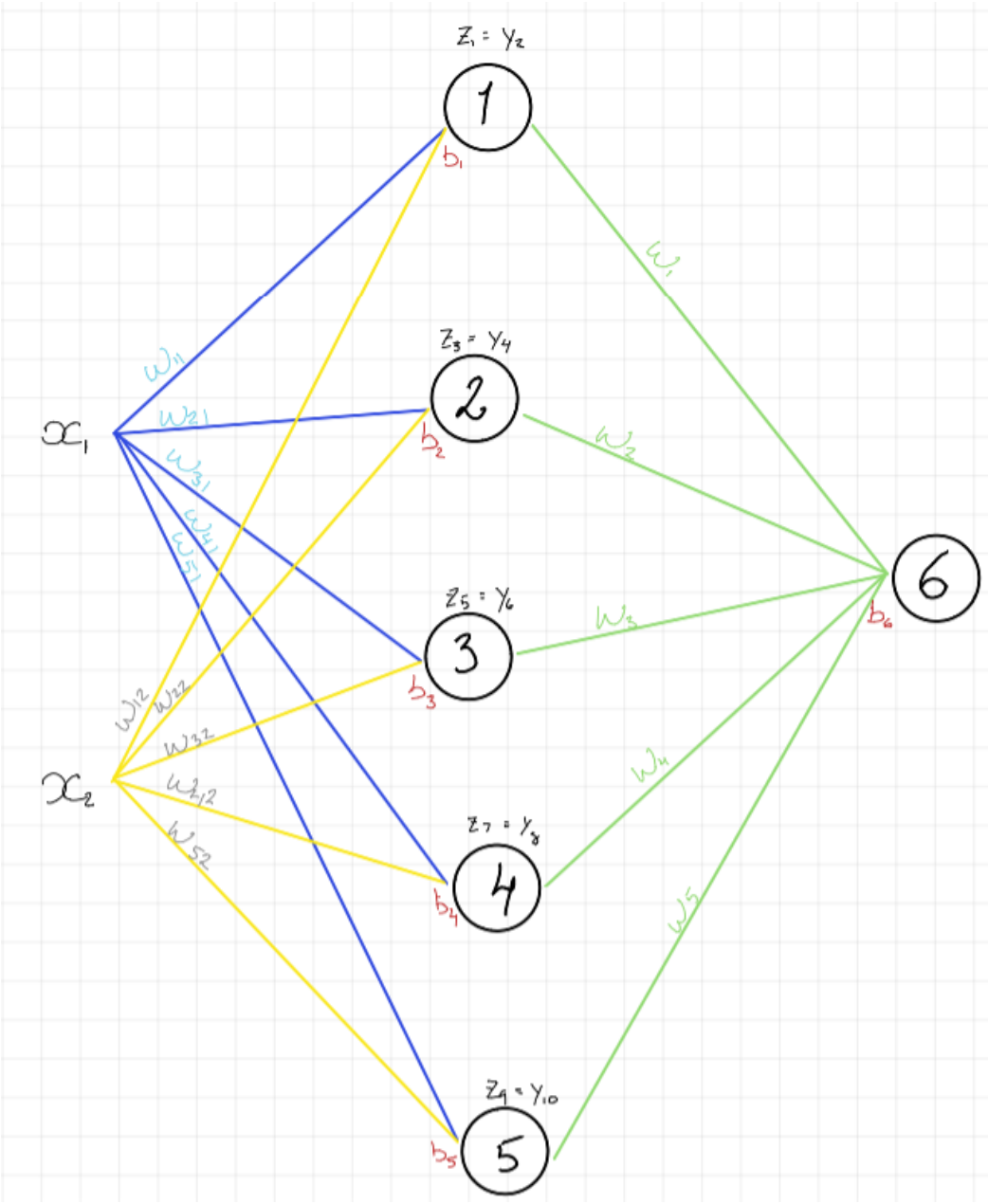


Figure 1: Neural network structure. The circled values represent the neurons. The weights (w) are indicated along each "arm" and the relevant bias (b) are indicated at each node (where the "arms" meet.

3 Gradients

In this section all the used equations of the gradients are given. First establishing the constant equations:

$$\begin{aligned}
\text{Activation Function for input: } & y_i = \sigma(z_i) = (1 + e^{-z_i})^{-1} \\
& \rightarrow \frac{\partial \sigma}{\partial z} = (1 - \sigma)\sigma \\
\text{Activation Function for output: } & \sigma_{i+1} = max(0, z_{i+1}) \\
& \rightarrow \frac{\partial \sigma}{\partial z} = 1 \\
\text{Total Loss Function: } & L(y, \hat{y}) = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2 \\
\text{Individual Loss Function: } & L_i = (y_i - \hat{y})^2 \\
\text{Weight: } & w_{i+1} = w_i - \eta \nabla L(w) \\
\text{Bias: } & b_{i+1} = b_i - \eta \nabla L(b)
\end{aligned}$$

As all the equations eventually are the same having only the subscripts varying - as such only the full maths will be shown for the first neuron:

$$x_1 x_2 \rightarrow (1) \rightarrow (6)$$

$$\begin{aligned}
z_1 &= w_{11}x_1 + w_{12}x_2 + b_1 && (1) \\
L_1 &= (y_1 - y)^2 && (2) \\
y_1 &= \sigma(z_1) && (3) \\
z_2 &= w_1 y_1 + b_6 && (4) \\
y_2 &= max(0, z_2) && (5) \\
&\textbf{Gradients} && (6) \\
\frac{\partial L_1}{\partial y_2} = \frac{\partial (y_2 - y)^2}{\partial y_2} &= 2(y_2 - y) && (7) \\
\frac{\partial L_1}{\partial w_1} = \frac{\partial L_1}{\partial y_2} * \frac{\partial y_2}{\partial z_2} * \frac{\partial z_2}{\partial w_1} &= 2(y_2 - y) * y_1 && (8) \\
\frac{\partial L_1}{\partial b_6} = \frac{\partial L_1}{\partial y_2} * \frac{\partial y_2}{\partial z_2} * \frac{\partial z_2}{\partial b_6} &= 2(y_2 - y) && (9) \\
\frac{\partial L_1}{\partial w_{11}} = \frac{\partial L_1}{\partial y_2} * \frac{\partial y_2}{\partial z_2} * \frac{\partial z_2}{\partial y_1} * \frac{\partial y_1}{\partial z_1} * \frac{\partial z_1}{\partial w_{11}} &= 2(y_2 - y) * w_1 * (1 - y_1)y_1 * x_1 && (10) \\
\frac{\partial L_1}{\partial w_{12}} = \frac{\partial L_1}{\partial y_2} * \frac{\partial y_2}{\partial z_2} * \frac{\partial z_2}{\partial y_1} * \frac{\partial y_1}{\partial z_1} * \frac{\partial z_1}{\partial w_{12}} &= 2(y_2 - y) * w_1 * (1 - y_1)y_1 * x_2 && (11) \\
\frac{\partial L_1}{\partial b_1} = \frac{\partial L_1}{\partial y_2} * \frac{\partial y_2}{\partial z_2} * \frac{\partial z_2}{\partial y_1} * \frac{\partial y_1}{\partial z_1} * \frac{\partial z_1}{\partial b_1} &= 2(y_2 - y) * w_1 * (1 - y_1)y_1 && (12) \\
& && (13)
\end{aligned}$$

The rest of the neuron equations will just be shown in their final equation:

$$x_1 x_2 \rightarrow \textcircled{2} \rightarrow \textcircled{6}$$

$$\begin{aligned}
z_3 &= w_{21}x_1 + w_{22}x_2 + b_2 \\
z_4 &= w_2 y_3 + b_7 \\
\nabla L(w_2) &= 2(y_4 - y) * y_3 \\
\nabla L(b_2) &= 2(y_4 - y) \\
\nabla L(w_{21}) &= 2(y_4 - y) * w_2 * (1 - y_3)y_3 * x_1 \\
\nabla L(w_{22}) &= 2(y_4 - y) * w_2 * (1 - y_3)y_3 * x_2 \\
\nabla L(b_7) &= 2(y_4 - y) * w_2 * (1 - y_3)y_3
\end{aligned}$$

$$x_1 x_2 \rightarrow \textcircled{3} \rightarrow \textcircled{6}$$

$$\begin{aligned}
z_5 &= w_{31}x_1 + w_{32}x_2 + b_3 \\
z_6 &= w_3 y_5 + b_8 \\
\nabla L(w_3) &= 2(y_6 - y) * y_5 \\
\nabla L(b_3) &= 2(y_6 - y) \\
\nabla L(w_{31}) &= 2(y_6 - y) * w_3 * (1 - y_5)y_5 * x_1 \\
\nabla L(w_{32}) &= 2(y_6 - y) * w_3 * (1 - y_5)y_5 * x_2 \\
\nabla L(b_8) &= 2(y_6 - y) * w_3 * (1 - y_5)y_5
\end{aligned}$$

$$x_1 x_2 \rightarrow \textcircled{4} \rightarrow \textcircled{6}$$

$$\begin{aligned}
z_7 &= w_{41}x_1 + w_{42}x_2 + b_4 \\
z_8 &= w_4 y_7 + b_9 \\
\nabla L(w_4) &= 2(y_8 - y) * y_7 \\
\nabla L(b_4) &= 2(y_8 - y) \\
\nabla L(w_{41}) &= 2(y_8 - y) * w_4 * (1 - y_7)y_7 * x_1 \\
\nabla L(w_{42}) &= 2(y_8 - y) * w_4 * (1 - y_7)y_7 * x_2 \\
\nabla L(b_9) &= 2(y_8 - y) * w_4 * (1 - y_7)y_7
\end{aligned}$$

$$x_1 x_2 \rightarrow \textcircled{5} \rightarrow \textcircled{6}$$

$$
\begin{aligned}
z_9 &= w_{51}x_1 + w_{52}x_2 + b_5 \\
z_{10} &= w_5 y_9 + b_{10} \\
\nabla L(w_5) &= 2(y_{10} - y) * y_9 \\
\nabla L(b_5) &= 2(y_{10} - y) \\
\nabla L(w_{51}) &= 2(y_{10} - y) * w_5 * (1 - y_9)y_9 * x_1 \\
\nabla L(w_{52}) &= 2(y_{10} - y) * w_5 * (1 - y_9)y_9 * x_2 \\
\nabla L(b_{10}) &= 2(y_{10} - y) * w_5 * (1 - y_9)y_9
\end{aligned}
$$

Step	Prediction	Loss	dLdb6	w11	w21	w31	w41	w51	w12	w22	w32	w42	w52	dLdw_11	dLdw_21	dLdw_31
1.0000	2.0315	0.9379	-1.9369	0.5000	0.4000	0.3000	0.5000	0.7000	0.4000	0.4000	0.8000	0.3000	0.5000	0.3829	0.6719	0.4466
2.0000	2.9588	4.1664	-4.0824	0.4617	0.3328	0.2553	0.4808	0.6768	0.3617	0.3328	0.7553	0.2808	0.4768	0.8276	1.2082	0.7566
3.0000	4.8198	4.7533	-4.3604	0.3790	0.2120	0.1797	0.4483	0.6459	0.2790	0.2120	0.6797	0.2483	0.4459	0.8509	0.9942	0.6179
4.0000	6.9274	4.2956	-4.1452	0.2939	0.1126	0.1179	0.4193	0.6245	0.1939	0.1126	0.6179	0.2193	0.4245	0.6034	0.6008	0.3581
5.0000	9.0116	3.9537	-3.9768	0.2335	0.0525	0.0821	0.4017	0.6138	0.1335	0.0525	0.5821	0.2017	0.4138	0.2246	0.2061	0.1125
6.0000	11.0750	3.7058	-3.8501	0.2111	0.0319	0.0708	0.3960	0.6109	0.1111	0.0319	0.5708	0.1960	0.4109	-0.2430	-0.2139	-0.1040
7.0000	13.1702	3.3481	-3.6596	0.2354	0.0533	0.0812	0.4013	0.6132	0.1354	0.0533	0.5812	0.2013	0.4132	-0.7357	-0.6264	-0.2705
8.0000	15.3168	2.8331	-3.3663	0.3089	0.1159	0.1083	0.4155	0.6184	0.2089	0.1159	0.6083	0.2155	0.4184	-1.0765	-0.8933	-0.3533
9.0000	17.4506	2.4006	-3.0988	0.4166	0.2052	0.1436	0.4348	0.6247	0.3166	0.2052	0.6436	0.2348	0.4247	-1.1019	-0.9288	-0.3595
10.0000	19.4834	2.3000	-3.0331	0.5268	0.2981	0.1796	0.4558	0.6308	0.4268	0.2981	0.6796	0.2558	0.4308	-0.9057	-0.8237	-0.3349

dLdw_41	dLdw_51	dLdw_12	dLdw_22	dLdw_32	dLdw_42	dLdw_52	dLdb_1	dLdb_2	dLdb_3	dLdb_4	dLdb_5	dLdw_1	dLdw_2	dLdw_3	dLdw_4	dLdw_5
0.1918	0.2323	0.3829	0.6719	0.4466	0.1918	0.2323	-0.2444	-0.4289	-0.2850	-0.1224	-0.1483	-0.3801	-0.8464	-1.1016	-1.6493	-1.7293
0.3255	0.3090	0.8276	1.2082	0.7566	0.3255	0.3090	-0.6791	-0.9915	-0.6209	-0.2671	-0.2536	-1.1149	-2.2767	-2.8141	-3.6307	-3.8025
0.2895	0.2138	0.8509	0.9942	0.6179	0.2895	0.2138	-0.9776	-1.1422	-0.7099	-0.3326	-0.2456	-1.6665	-2.9833	-3.4574	-4.0085	-4.1706
0.1764	0.1073	0.6034	0.6008	0.3581	0.1764	0.1073	-1.1554	-1.1505	-0.6858	-0.3378	-0.2055	-2.0057	-3.1397	-3.5456	-3.8934	-4.0256
0.0568	0.0291	0.2246	0.2061	0.1125	0.0568	0.0291	-1.2901	-1.1842	-0.6465	-0.3264	-0.1670	-2.2276	-3.1600	-3.5485	-3.7883	-3.8984
-0.0533	-0.0231	-0.2430	-0.2139	-0.1040	-0.0533	-0.0231	-1.3960	-1.2286	-0.5976	-0.3063	-0.1329	-2.3881	-3.1482	-3.5318	-3.7052	-3.7975
-0.1417	-0.0527	-0.7357	-0.6264	-0.2705	-0.1417	-0.0527	-1.4089	-1.1995	-0.5180	-0.2714	-0.1009	-2.4997	-3.0789	-3.4292	-3.5506	-3.6250
-0.1930	-0.0626	-1.0765	-0.8933	-0.3533	-0.1930	-0.0626	-1.2368	-1.0263	-0.4059	-0.2218	-0.0720	-2.5665	-2.9413	-3.2112	-3.2887	-3.3444
-0.2098	-0.0608	-1.1019	-0.9288	-0.3595	-0.2098	-0.0608	-0.9042	-0.7622	-0.2950	-0.1721	-0.0499	-2.6330	-2.8254	-2.9993	-3.0449	-3.0850
-0.2125	-0.0564	-0.9057	-0.8237	-0.3349	-0.2125	-0.0564	-0.5781	-0.5258	-0.2137	-0.1356	-0.0360	-2.7862	-2.8666	-2.9682	-2.9945	-3.0240

Figure 2: Full table of results for $\eta = 0.1$, displaying outputs from each neuron

4 Results

For how the network runs please refer to the code.

Now to the results - these are tabulated results as the assignment expressed. Due to the large size of each table, they have had to be split in two to encompass all the desired variables: weights, gradients, outputs and loss.

The first result is in Fig.2 where $\eta = 0.1$. It can be seen here that while the initial loss prediction is low and then jumps up, it does trend towards a declining value. Where as for the other situation of $\eta > 0.1 \rightarrow \eta = 0.6$, there is an unevenness to the progression of the loss values and they tend to become larger.

Step	Prediction	Loss	dLdb6	w11	w21	w31	w41	w51	w12	w22	w32	w42	w52	dLdw_11	dLdw_21	dLdw_31
1.0000	2.0315	0.9379	-1.9369	0.5000	0.4000	0.3000	0.5000	0.7000	0.4000	0.4000	0.8000	0.3000	0.5000	0.3829	0.6719	0.4466
2.0000	6.8139	3.2902	3.6278	0.2703	-0.0032	0.0320	0.3849	0.5606	0.1703	-0.0032	0.5320	0.1849	0.3606	-1.0940	-1.0683	-0.8400
3.0000	-2.0678	82.2245	-18.1355	0.9267	0.6378	0.5360	0.6915	0.7876	0.8267	0.6378	1.0360	0.4915	0.5876	0.2633	-1.1032	-1.9509
4.0000	27.8670	355.9622	37.7339	0.7687	1.2997	1.7066	1.2740	0.9169	0.6687	1.2997	2.2066	1.0740	0.7169	-6.0847	-13.0246	-16.1121
5.0000	-12.7750	565.2513	-47.5500	4.4195	9.1145	11.3738	22.4768	6.7376	4.3195	9.1145	11.8738	22.2768	6.5376	-0.0064	-0.0000	0.0000
6.0000	15.6711	7.1347	5.3422	4.4233	9.1145	11.3738	22.4768	6.7411	4.3233	9.1145	11.8738	22.2768	6.5411	-0.0146	-0.0000	0.0000
7.0000	7.0825	62.6866	-15.8350	4.4320	9.1145	11.3738	22.4768	6.7797	4.3320	9.1145	11.8738	22.2768	6.5797	2.5239	0.0136	-0.0482
8.0000	19.2389	5.0125	4.4777	2.9177	9.1063	11.4028	22.4768	-7.3382	2.8177	9.1063	11.9028	22.2768	-7.5382	-0.0973	-0.2909	0.7756
9.0000	7.7092	127.4832	-22.5817	2.9761	9.2809	10.9374	26.5341	-7.3382	2.8761	9.2809	11.4374	26.3341	-7.5382	5.6827	0.0249	0.0006
10.0000	62.1147	1690.4156	82.2293	-0.4335	9.2659	10.9370	26.5341	-7.3382	-0.5335	9.2659	11.4370	26.3341	-7.5382	-0.0003	0.0010	0.0001

dLdw_41	dLdw_51	dLdw_12	dLdw_22	dLdw_32	dLdw_42	dLdw_52	dLdb_1	dLdb_2	dLdb_3	dLdb_4	dLdb_5	dLdw_1	dLdw_2	dLdw_3	dLdw_4	dLdw_5
0.1918	0.2323	0.3829	0.6719	0.4466	0.1918	0.2323	-0.2444	-0.4289	-0.2850	-0.1224	-0.1483	-0.3801	-0.8464	-1.1016	-1.6493	-1.7293
-0.5109	-0.3783	-1.0940	-1.0683	-0.8400	-0.5109	-0.3783	0.8978	0.8767	0.6894	0.4193	0.3105	1.4645	2.8293	2.9571	3.3202	3.4501
-0.9710	-0.2155	0.2633	-1.1032	-1.9509	-0.9710	-0.2155	-0.3025	1.2674	2.2414	1.1155	0.2475	-2.3227	-7.3695	-10.8099	-15.5459	-16.9996
-35.3379	-9.7012	-6.0847	-13.0246	-16.1121	-35.3379	-9.7012	11.6512	24.9401	30.8523	67.6670	18.5763	10.4390	7.5442	6.1767	27.0298	35.7677
-0.0000	-0.0058	-0.0064	-0.0000	0.0000	-0.0000	-0.0058	0.0365	0.0000	-0.0000	0.0000	0.0334	-0.0077	-0.0000	-0.0000	-0.0000	-0.0029
-0.0000	-0.0644	-0.0146	-0.0000	0.0000	-0.0000	-0.0644	-0.0837	-0.0000	0.0000	-0.0000	-0.3699	0.0178	0.0000	0.0000	0.0000	0.0324
0.0000	23.5299	2.5239	0.0136	-0.0482	0.0000	23.5299	4.8329	0.0260	-0.0923	0.0000	45.0564	-1.0988	-0.0662	-0.0408	-0.0000	-7.0775
-6.7622	-0.0000	-0.0973	-0.2909	0.7756	-6.7622	-0.0000	-0.1118	-0.3342	0.8911	-7.7692	-0.0000	0.0277	3.1306	4.0474	2.7928	0.0000
0.0000	0.0000	5.6827	0.0249	0.0006	0.0000	0.0000	4.6635	0.0204	0.0005	0.0000	0.0000	-1.2064	-22.5725	-22.5779	-22.5817	-0.0000
0.0000	-0.0000	-0.0003	0.0010	0.0001	0.0000	-0.0000	-0.0002	0.0006	0.0001	0.0000	-0.0000	0.0000	82.2293	82.2293	82.2293	0.0000

Figure 3: Full table of results for $\eta = 0.6$, displaying outputs from each neuron